Fiscal Year:	FY 2024	**Task Last Updated:**	FY 04/01/2024
PI Name:	Juran, Cassandra Ph.D.		
Project Title:	Sustained Effects of Spaceflight on Anemia and Severity of Effects Dependency on Age		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Animal Biology: Vertebrate		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	(1) Cell Culture		
PI Email:	cassandra.m.juran@nasa.gov	**Fax:**	FY
PI Organization Type:	NASA CENTER	**Phone:**	954-275-0136
Organization Name:	NASA Ames Research Center		
PI Address 1:	Blue Marble Space Institute of Science Affiliate Research Scientist (BMSIS)		
PI Address 2:	MS 236-7		
PI Web Page:			
City:	Moffett Field	**State:**	CA
Zip Code:	94035	**Congressional District:**	18
Comments:			
Project Type:	Ground,NASA GeneLab	**Solicitation / Funding Source:**	2021 Space Biology NNH21ZDA001N-SBAS E.11: Animal Studies
Start Date:	04/01/2023	**End Date:**	04/03/2025
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA ARC
Contact Monitor:	Griko, Yuri	**Contact Phone:**	650-604-0519
Contact Email:	Yuri.V.Griko@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: End date changed to 04/03/2025 per F. Hernandez/ARC. Previous end date was 03/31/2024 (Ed., 3/15/24).		
Key Personnel Changes/Previous PI:	No change.		
COI Name (Institution):	Almeida, Eduardo Ph.D. (NASA Ames Research Center)		
Grant/Contract No.:	Internal Project		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Anemia is a significant risk to the wellbeing of astronauts. Recent studies have demonstrated that these risks do not normalize over long duration missions, as was once thought, and suppressed red blood cell numbers are retained long after return to Earth. Additionally, age influences severity of red blood cell depletion on Earth and would likely become a significant contributor to space anemia as longer duration missions to Moon, Mars, or permanent colonization missions are considered. Anemia is a condition which can cause fatigue and physical weakness; both are detrimental to an astronaut's ability to conduct tasks requiring stamina, concentration, and precision. Additionally, long term untreated anemia corresponds to heart failure and a compromised immune system. As space anemia is associated with so many risks, there has been significant research into its cause. Recent findings determined hemolysis was a major contributor to space anemia, showing a 54% increase in hemolysis markers in expiration and blood sampling; however, this study did not probe the molecular mechanisms driving the observed hemolysis. The NASA GeneLab program has recently released a series of four single cell RNA-sequencing datasets that our proposal seeks to consult to better understand how spaceflight and aging impacts blood cell production and destruction throughout the red blood cell's life cycle. The GeneLab datasets to be interrogated include three tissue types: 1) the source of new red blood cells (bone marrow); 2) circulating blood samples (peripheral bloodmononuclear cells: PBMCs); and 3) the site of terminal or dysfunctional red blood cell removal (spleen) collected from two age groups, one equating to young adult humans and the other associated with the average age of the astronaut population. These data analyses are uniquely significant as the datasets include tissues pertinent for the entire life cycle of a red blood cell, allowing mechanistic explanation of suppressed red blood cell development (erythropoiesis) and increased rate of hemolysis. Future work expanding on this one-year GeneLab Analysis Investigation could probe other hematopoietic/immune cell lineages using methodology similar to those developed for erythropoiesis or could use the omics regulation findings of this proposal to investigate molecular countermeasures for space anemia. The analyses conducted herein, and methods generated, could also be impactful for future Space Biology investigations at single cell resolution. One such application is analysis of the recently completed Rodent Research 10 mission, in which single cell bone marrow flushes were cryopreserved in orbit and processed for single cell sequencing on the ground. These analyses are critical as they will define spaceflight risks to the oxygen transport system of the body and characterize the duration of molecular changes after return to the ground. The main goal of this proposal is to understand the molecular causes of anemia to devise countermeasures to protect our astronaut population during and after return from spaceflight duties.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Anemia is a significant risk to the wellbeing of astronauts. Recent studies have demonstrated that these risks do not normalize over long duration missions, as was once thought, and suppressed red blood cell (RBC) numbers are retained long after return to Earth. Additionally, age influences severity of RBC depletion on Earth and would likely become a significant contributor to space anemia as longer duration missions to Moon, Mars or permanent colonization missions are considered. Anemia is a condition which can cause fatigue and physical weakness, both detrimental to an astronaut's ability to conduct tasks requiring stamina, concentration and precision. Additionally, long term untreated anemia corresponds to heart failure and compromised immune system. As space anemia is associated with so many risks, there has been significant research into its cause. Recent findings determined hemolysis was a major contributor to space anemia, showing a 54% increase in hemolysis markers in expiration and blood sampling, however, this study did not probe the molecular mechanisms driving the observed hemolysis. The NASA GeneLab data repository released a series of four single cell RNA-sequencing datasets that our proposal has utilized to develop a molecular regulation understanding of how spaceflight and aging impacts blood cell production and destruction throughout the red blood cell's lifecycle. The GeneLab datasets to be interrogated include three tissue types 1) the source of new red blood cells (bone marrow), 2) circulating blood samples (PBMCs), and 3) the site of terminal or dysfunctional red blood cell removal (spleen) collected from two two age groups, one equating to young adult humans and the other associated with the average age of the astronaut population. These data analyses are uniquely significant as the datasets include tissues pertinent for the entire life cycle of a red blood cell, allowing mechanistic explanation of suppressed red blood cell development (erythropoiesis) and increased rate of hemolysis. Additionally, our analysis has expanded to include hematopoietic/immune cell lineages co-regulation of eryhtorpoiesis. The analyses conducted herein, and methods generated, are impactful for future Space Biology investigations at single cell resolution and may lead to new innovation in understanding anemia, the diseases causes and potential therapies, here on Earth. The main goal of this proposal has been to understand the molecular causes of anemia such to devise countermeasures to protect our astronaut population during and after return from spaceflight duties.
Task Progress:	Our team has generated individual tissue type single cell profiling for circulating blood, the spleen, femur, and humerus bone marrow. From these analyses, we excised the erythroid lineage cells and conducted analysis determining the key regulatory transcriptomic signature for the red blood cell lifecycle across the tissue listed above. In effect, we created a molecular roadmap of the red blood cell (RBC) lifecycle. We then compared the roadmaps generated for the four experimental conditions (mice age 12 weeks at launch, and mice 29 weeks at launch, both flight and ground control), generating the differential gene expression atlas for spaceflight dysregulation of erythropoiesis and the effect of age on this dysregulation. We are in final preparation of a manuscript describing the above science; however, after our initial evaluation of findings, we decided to expand our analytics to include cell-cell communication and co-regulation mapping. By including this analysis, we can provide the community a more detailed and holistic explanation of space anemia. In the bone marrow, for example, t-cells closely communicate with common myeloid progenitors, but not with later RBC differentiation lineages. And, in the spleen, our data show a significant shift toward RBCs marked for phagocytosis in the spleen, implying either an increased rate of malformed RBCs being produced or an immune system bias toward their removal. We further analyzed the data from the Humerus, to produce a second study in which we defined the biologic diversity of individual animals using high resolution single cell RNA sequencing data. This manuscript, currently in late phase generation, evaluates the statistical significance of biologic divergence of individual animals on single cell analyses with 4 biologic replicates per condition, and presents this review, for the first time, to the community.
Bibliography Type:	Description: (Last Updated: 04/22/2024)

Abstracts for Journals and Proceedings	Brown KM, Mistry E, Almeida EAC, Juran CM. "Spaceflight effects on erythropoiesis and its dependency on age." 39th Annual Meeting of the American Society for Gravitational and Space Research, Washington, DC, November 13-18, 2023. Abstracts. 39th Annual Meeting of the American Society for Gravitational and Space Research, Washington, DC, November 13-18, 2023. , Nov-2023